# 3. Reading and Writing Data in Pandas

In Pandas, data can come from:

- CSV files
- Excel files
- JSON files
- SQL Databases

Pandas can also save data into:

- CSV
- Excel

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**First Step: Import Pandas**

import pandas as pd

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**1. read_csv()**

read_csv() is used to read CSV files.

CSV = Comma Separated Values

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**What is CSV File?**

Example:

students.csv

| Name | Age | Marks |
|---|---|---|
| **Rahul** | 20 | 85 |
| **Aman** | 21 | 90 |
| **Priya** | 19 | 88 |

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**Step-by-Step Example**

### Step 1: Create CSV File

Save this as:

```
students.csv
```

Data inside file:

```
Name,Age,Marks
Rahul,20,85
Aman,21,90
Priya,19,88
```

### Step 2: Read CSV File

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df)
```

### Output

| | Name | Age | Marks |
|---|---|---|---|
| **0** | Rahul | 20 | 85 |
| **1** | Aman | 21 | 90 |
| **2** | Priya | 19 | 88 |

### Understanding the Code

### Step 1

```
pd.read_csv()
```

Reads CSV file.

**Step 2**

```
df
```

Stores data in DataFrame.

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**Useful Options in read_csv()**

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**Read First 2 Rows**

```
df = pd.read_csv("students.csv", nrows=2)

print(df)
```

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**Select Specific Columns**

```
df = pd.read_csv(
    "students.csv",
    usecols=["Name", "Marks"]
)

print(df)
```

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**2. read_excel()**

Used to read Excel files.

Supported formats:

- .xlsx
- .xls

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**Install Excel Support**

```
pip install openpyxl
```

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**Example Excel File**

students.xlsx

| Name | Age | Marks |
|---|---|---|
| **Rahul** | 20 | 85 |
| **Aman** | 21 | 90 |

---

**Read Excel File**

```
import pandas as pd

df = pd.read_excel("students.xlsx")

print(df)
```

---

**Output**

| | Name | Age | Marks |
|---|---|---|---|
| **0** | Rahul | 20 | 85 |
| **1** | Aman | 21 | 90 |

---

**Read Specific Sheet**

```
df = pd.read_excel(
   "students.xlsx",
   sheet_name="Sheet1"
)

print(df)
```

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**3. read_json()**

Used to read JSON data.

JSON = JavaScript Object Notation

Mostly used in:

- APIs
- Web applications

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**Example JSON File**

students.json

```
[
 {"Name":"Rahul","Age":20,"Marks":85},
 {"Name":"Aman","Age":21,"Marks":90}
]
```

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**Read JSON File**

```
import pandas as pd

df = pd.read_json("students.json")

print(df)
```

---

**Output**

| | Name | Age | Marks |
|---|---|---|---|
| **0** | Rahul | 20 | 85 |
| **1** | Aman | 21 | 90 |

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**4. read_sql()**

Used to read data from database.

Works with:

- MySQL
- PostgreSQL

- SQLite
- SQL Server

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**Install Required Packages**

```
pip install sqlalchemy pymysql
```

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**MySQL Example**

Suppose database name:

```
school
```

Table:

```
students
```

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**Database Table**

| id | name | marks |
|---|---|---|
| **1** | Rahul | 85 |
| **2** | Aman | 90 |

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**Step-by-Step Example**

```
from sqlalchemy import create_engine
import pandas as pd

engine = create_engine(
    "mysql+pymysql://root:password@localhost/school"
)

df = pd.read_sql(
    "SELECT * FROM students",
    engine
)

print(df)
```

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**Understanding the Code**

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**create_engine()**

Connects Python with database.

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**read_sql()**

Runs SQL query and converts result into DataFrame.

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**5. to_csv()**

Used to save DataFrame into CSV file.

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**Example**

```
import pandas as pd

data = {
    "Name": ["Rahul", "Aman"],
    "Marks": [85, 90]
}

df = pd.DataFrame(data)

df.to_csv("output.csv")
```

---

**Output File**

output.csv

Contains:

```
,Name,Marks
0,Rahul,85
1,Aman,90
```

---

**Remove Index Column**

```
df.to_csv(
    "output.csv",
    index=False
)
```

---

**Better Output**

Name,Marks
Rahul,85
Aman,90

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**6. to_excel()**

Used to save DataFrame into Excel file.

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**Install Excel Support**

```
pip install openpyxl
```

---

**Example**

```
import pandas as pd

data = {
    "Name": ["Rahul", "Aman"],
    "Marks": [85, 90]
}

df = pd.DataFrame(data)

df.to_excel(
    "students_output.xlsx",
    index=False
)
```

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**Output**

Creates:

```
students_output.xlsx
```

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**Real-Life Workflow Example**

Suppose company has:

- Sales CSV files
- Employee Excel files
- Customer JSON APIs
- SQL database

Pandas can:

1. Read all data
2. Combine data
3. Analyze reports
4. Export final report

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**Full Example (Complete Flow)**

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**Step 1: Read CSV**

```
df = pd.read_csv("students.csv")
```

---

**Step 2: Show Data**

```
print(df)
```

---

**Step 3: Add New Column**

```
df["Result"] = "Pass"
```

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**Step 4: Save New File**

```
df.to_excel(
    "final_report.xlsx",
    index=False
)
```

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**Summary Table**

| Function | Purpose |
|---|---|
| **read_csv()** | Read CSV file |
| **read_excel()** | Read Excel file |
| **read_json()** | Read JSON file |
| **read_sql()** | Read database table |
| **to_csv()** | Save CSV file |
| **to_excel()** | Save Excel file |

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**Important Notes**

| File Type | Function |
|---|---|
| **CSV** | read_csv() |
| **Excel** | read_excel() |
| **JSON** | read_json() |
| **Database** | read_sql() |